

COAL LAB#7:

NAME: ARSAM ALI

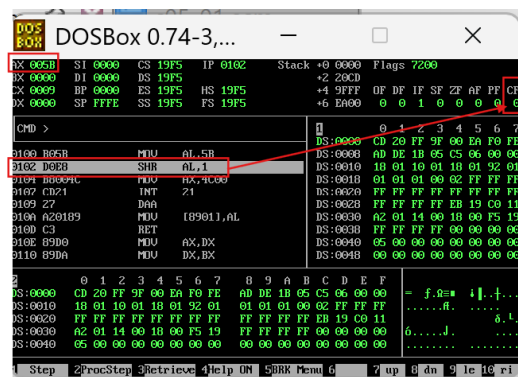
ROLL-NO: 23P-0038

PROGRAM/SECTION: AI/4-A

Task 1: SHR SHIFT RIGHT:

Code :

```
1 [org 0x0100]
2
3 mov al,01011011b
4
5 shr al,1
6
7 mov ax,0x4c00;
8 int 0x21;
9
```



DOSBox 0.74-3,...

AX: 005B SI: 0000 CS: 19F5 IP: 0102 Stack: +0 0000 Flags: 7200

DX: 0000 DI: 0000 DS: 19F5 +2 20CD

CX: 0000 BP: 0000 ES: 19F5 HS: 19F5 +4 9FFF OF DF IF SF ZF AF CF

FX: 0000 SP: FFFE SS: 19F5 FS: 19F5 +6 EA00 0 0 1 0 0 0 0 0

CF: 0

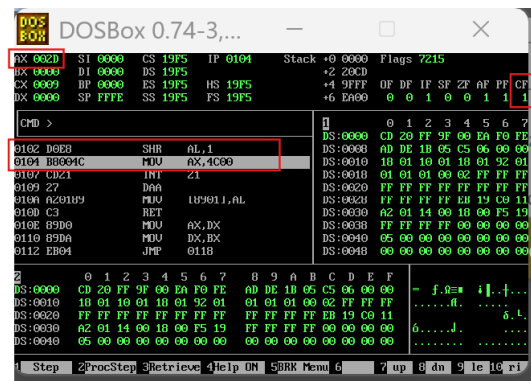
Command window: CMD > SHR AL,1

Register window: AX: 005B, AL: 5B, CF: 0

Memory window: 0000: 00 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00

Before the shift right we can see:

- Ax store hex of given binary of given value
- cf is 0



DOSBox 0.74-3,...

AX: 002B SI: 0000 CS: 19F5 IP: 0104 Stack: +0 0000 Flags: 7215

DX: 0000 DI: 0000 DS: 19F5 +2 20CD

CX: 0000 BP: 0000 ES: 19F5 HS: 19F5 +4 9FFF OF DF IF SF ZF AF CF

FX: 0000 SP: FFFE SS: 19F5 FS: 19F5 +6 EA00 0 0 1 0 0 1 1 1

CF: 1

Command window: CMD > SHR AL,1

Register window: AX: 002B, AL: 2B, CF: 1

Memory window: 0000: 00 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00

After the shift we can see:

- cf stores 1 as LSB was 1
- value of ax has been changed from 5B->2B

HEX	2B
DEC	43
OCT	53
BIN	0010 1011

FROM CODE we see:

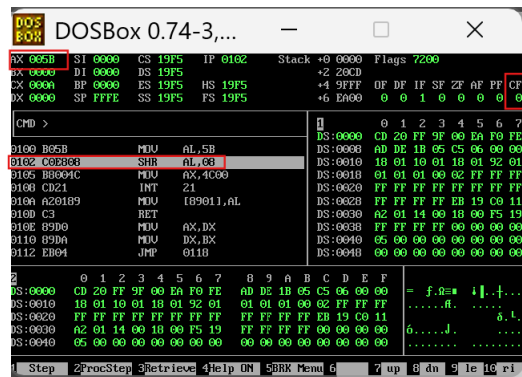
initially the value was 01011011(5B)

after shift value became 0010 1011(0 added to MSB and LSB sent to cf)

Now if we shifted right 8 bits on same value:

Code:

```
1 [org 0x0100]
2
3 mov al,01011011b
4
5 shr al,8
6
7 mov ax,0x4c00;
8 int 0x21;
9
```



DOSBox 0.74-3,...

Register	Value
AX	005B
SI	0000
DI	0000
BP	0000
SP	FFFE
CS	19F5
DS	19F5
ES	19F5
FS	19F5

Stack: +0 0000, +2 20CD, +4 9FFF, +6 EA00

Flags: 7200

CF: 0

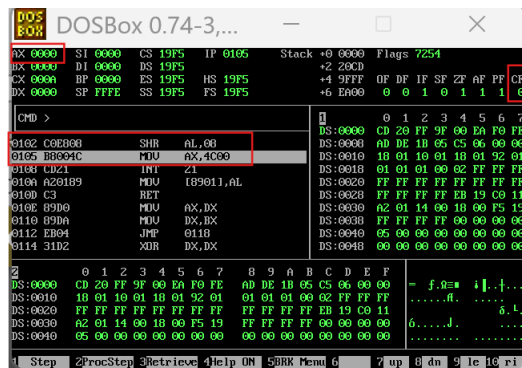
Command Line: CMD >

Assembly Code:

```
0100 B05B MOV AL,5B
0102 C0B0B0 SHR AL,08
0105 B0004C MOV AX,4C00
0108 CD21 INT 21
010A A20109 MOV EB,0109
010D C3 RET
010E B9D0 MOV AX,DX
0110 B9D0 MOV DX,BX
0112 EB04 JMP 011B
```

Ax initially stores the given value 5B

And CF is 0



DOSBox 0.74-3,...

Register	Value
AX	0000
SI	0000
DI	0000
BP	0000
SP	FFFE
CS	19F5
DS	19F5
ES	19F5
FS	19F5

Stack: +0 0000, +2 20CD, +4 9FFF, +6 EA00

Flags: 7254

CF: 0

Command Line: CMD >

Assembly Code:

```
0102 C0B0B0 SHR AL,08
0105 B0004C MOV AX,4C00
0108 CD21 INT 21
010A A20109 MOV EB,0109
010D C3 RET
010E B9D0 MOV AX,DX
0110 B9D0 MOV DX,BX
0112 EB04 JMP 011B
0114 31D2 XDR DX,DX
```

Now Ax has turned 0 as al register stores 8 bits of data

With each traversal MSB as changed to 0

So with 8 shifts 0 is added to left side 8 times making AX 0.

Task 2 :Shift Logical Left (SHL):

```

1 [org 0x0100]
2
3 mov ax,0101101110110111b
4
5 shl ax,8
6
7 mov ax,0x4c00;
8 int 0x21;
9

```

DOSBox 0.74-3,...

AX: 5BB7	SI: 0000	CS: 19F5	IP: 0103	Stack: +0 0000	Flags: 7200
BX: 0000	DI: 0000	DS: 19F5		+2 20CD	
CX: 0000	BP: 0000	ES: 19F5	HS: 19F5	+4 9FFF	OF DF IF SF ZF AF CF 0
DX: 0000	SP: FFFE	SS: 19F5	FS: 19F5	+6 E000	0 0 1 0 0 0 0 0

CMD > |

0100 B8B75B	MOV	AX,5BB7	DS:0000	CD 20 FF 9F 00 EA F0 FE
0103 C1E000	SHL	AX,08	DS:0010	AD DE 1B 05 C5 06 00 00
0104 B8004C	MOV	AX,4C00	DS:0018	01 01 01 00 02 FF FF FF
0109 CD21	INT	21	DS:0020	FF FF FF FF FF FF FF FF
010B 019C3B9	ADD	[BP+3+BX+DI],CX	DS:0028	FF FF FF FF EB 19 C0 11
010F D09DAEB	ROR	B:[EBDI+BX+DI],1	DS:0030	A2 01 14 00 10 00 F5 19
0113 0431	ADD	AL,31	DS:0038	FF FF FF FF 00 00 00 00
0115 D2	DB	D2	DS:0040	05 00 00 00 00 00 00 00
0116 31C9	XOR	AX,AX	DS:0048	00 00 00 00 00 00 00 00

- Before shift AX stores 5BB7(0101101110110111)
- CF is 0

DOSBox 0.74-3,...

AX: B700	SI: 0000	CS: 19F5	IP: 0106	Stack: +0 0000	Flags: 7095
BX: 0000	DI: 0000	DS: 19F5		+2 20CD	
CX: 0000	BP: 0000	ES: 19F5	HS: 19F5	+4 9FFF	OF DF IF SF ZF AF CF 1
DX: 0000	SP: FFFE	SS: 19F5	FS: 19F5	+6 E000	1 0 1 1 0 1 0 1

CMD > |

0103 C1E000	SHL	AX,08	DS:0000	CD 20 FF 9F 00 EA F0 FE
0104 B8004C	MOV	AX,4C00	DS:0010	AD DE 1B 05 C5 06 00 00
0109 CD21	INT	21	DS:0018	01 01 01 00 02 FF FF FF
010B 019C3B9	ADD	[BP+3+BX+DI],CX	DS:0020	FF FF FF FF FF FF FF FF
010F D09DAEB	ROR	B:[EBDI+BX+DI],1	DS:0028	FF FF FF FF EB 19 C0 11
0113 0431	ADD	AL,31	DS:0030	A2 01 14 00 10 00 F5 19
0115 D2	DB	D2	DS:0038	FF FF FF FF 00 00 00 00
0116 31C9	XOR	AX,AX	DS:0040	05 00 00 00 00 00 00 00
011B 8956E4	MOV	[BP-1C],DX	DS:0048	00 00 00 00 00 00 00 00

After the shift Ax stores B700 (1011011100000000)

From above binary we can see 8 zeros are added from the Left and carry flag stores value of 8th bit (bolded above)

NOW, if we shift 16 bit left:

```
1 [org 0x0100]
2
3 mov ax,0101101110110111b
4
5 shl ax,16
6
7 mov ax,0x4c00;
8 int 0x21;
9
```

Before shift:

The screenshot shows the DOSBox 0.74-3 interface. The assembly window displays the following code:

```
0100 B8B75B MOV AX,5BB7
0103 C1E010 SHL AX,10
0106 B8004C MOV AX,4C00
0109 CD21 INT Z1
010B 0109C3B9 ADD [09C3+BX+DI],CX
010F D093DAEB ROR B,[EBDA+BX+DI],1
0113 0431 ADD AL,31
0115 D2 DB D2
0116 31C9 XOR AX,AX
```

The memory dump shows the state of memory before the shift operation. The AX register is highlighted in red, showing the value 5BB7 (0101101110110111b).

After shift:

The screenshot shows the DOSBox 0.74-3 interface after the shift operation. The assembly window displays the following code:

```
0103 C1E010 SHL AX,10
0106 B8004C MOV AX,4C00
0109 CD21 INT Z1
010B 0109C3B9 ADD [09C3+BX+DI],CX
010F D093DAEB ROR B,[EBDA+BX+DI],1
0113 0431 ADD AL,31
0115 D2 DB D2
0116 31C9 XOR AX,AX
0118 8956E4 MOV [BP-1C],DX
```

The memory dump shows the state of memory after the shift operation. The AX register is highlighted in red, showing the value 0000 (0000000000000000b).

From above screenshot we can see AX initially stores 5BB7 (0101101110110111) which 16 times shifts left (adding zero to the LSB) and changing the value to 0 as AX

SHIFT ARITHMETIC RIGHT (SAR):

```

1 [org 0x0100]
2
3 mov al,10110101b
4
5 sar al,1
6
7 mov ax,0x4c00;
8 int 0x21;
9

```

Before shifting:

Register	Value
AX	00B5
BX	0000
CX	0000
DX	0000
SI	0000
DI	0000
BP	0000
SP	FFFE
CS	19F5
DS	19F5
ES	19F5
SS	19F5
FS	19F5

Address	Instruction	Comment
0100	B0B5	MOV AL,B5
0102	D0FB	SAR AL,1
0104	B8004C	MOV AX,4C00
0107	CD21	INT 21
0109	Z7	DAA
010A	A201B9	MOV [0901],AL
010D	C3	RET
010E	89D0	MOV AX,DX
0110	89DA	MOV DX,BX

After shifting:

Register	Value
AX	00DA
BX	0000
CX	0000
DX	0000
SI	0000
DI	0000
BP	0000
SP	FFFE
CS	19F5
DS	19F5
ES	19F5
SS	19F5
FS	19F5

Address	Instruction	Comment
0102	D0FB	SAR AL,1
0104	B8004C	MOV AX,4C00
0107	CD21	INT 21
0109	Z7	DAA
010A	A201B9	MOV [0901],AL
010D	C3	RET
010E	89D0	MOV AX,DX
0110	89DA	MOV DX,BX
0112	EB04	JMP 0118

- AX has changed from B5(10110101) to DA(11011010)
- Carry flag changed from 0 to 1
- MSB remained the same as before

We see that the LSB has been sent to carry flag

MSB has been copied to the second MSB

SHIFT ARITHMETIC RIGHT (SAR): shifting 8 bit data

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | sar al,8
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:

```
DOSBox 0.74-3,...
AX 00B5 SI 0000 CS 19F5 IP 0102 Stack +0 0000 Flags 7200
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 0000 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

CMD >
0100 B0B5 MOV AL,B5
0102 C0F808 SAR AL,08
0105 B004C MOV AX,4C00
0106 CD21 INT 21
010A A201B9 MOV [8901],AL
010D C3 RET
010E B9D0 MOV AX,DX
0110 B9DA MOV DX,BX
0112 EB04 JMP 0118

DS:0000 0 1 2 3 4 5 6 7
DS:0000 CD 20 FF 9F 00 EA F0 FE
DS:0010 AD DE 1B 05 C5 06 00 00
DS:0018 18 01 10 01 18 01 92 01
DS:0020 FF FF FF FF FF FF FF FF
DS:0028 FF FF FF FF FF FF FF FF
DS:0030 A2 01 14 00 18 00 F5 19
DS:0038 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00
DS:0048 00 00 00 00 00 00 00 00
```

After shifting:

```
DOSBox 0.74-3,...
AX 00FF SI 0000 CS 19F5 IP 0105 Stack +0 0000 Flags 7295
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 0000 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 1 1

CMD >
0102 C0F808 SAR AL,08
0105 B004C MOV AX,4C00
0106 CD21 INT 21
010A A201B9 MOV [8901],AL
010D C3 RET
010E B9D0 MOV AX,DX
0110 B9DA MOV DX,BX
0112 EB04 JMP 0118
0114 51D2 XDB DX,BX

DS:0000 0 1 2 3 4 5 6 7
DS:0000 CD 20 FF 9F 00 EA F0 FE
DS:0010 AD DE 1B 05 C5 06 00 00
DS:0018 18 01 10 01 18 01 92 01
DS:0020 FF FF FF FF FF FF FF FF
DS:0028 FF FF FF FF FF FF FF FF
DS:0030 A2 01 14 00 18 00 F5 19
DS:0038 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00
DS:0048 00 00 00 00 00 00 00 00
```

- AX has changed from B5(10110101) to FF(11111111)
- Carry flag changed from 0 to 1

This is because the MSB has been copied to the second MSB with each shift to the right

Rotate Right (ROR): ROTATING 1 BIT DATA:

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | ROR al,1
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:

AX	SI	DI	BP	SP	CS	DS	ES	FS	IP	Stack	Flags
00B5	0000	0000	0000	FFFE	19F5	19F5	19F5	19F5	0102	+0 0000	7200

0100	0101	0102	0103	0104	0105	0106	0107	0108	0109	010A	010B	010C	010D	010E	010F
B0B5	MOV	AL,B5		B0B6	ROR	AL,1		B0B7	MOV	AX,4C00		B0B8	INT	21	

After shifting:

AX	SI	DI	BP	SP	CS	DS	ES	FS	IP	Stack	Flags
00DA	0000	0000	0000	FFFE	19F5	19F5	19F5	19F5	0104	+0 0000	7201

0102	0103	0104	0105	0106	0107	0108	0109	010A	010B	010C	010D	010E	010F	0110	0111
D0CB	ROR	AL,1		B0B6	MOV	AX,4C00		B0B7	INT	21		B0B8	DAA		

- AX has changed from B5(10110101) to DA(11011010)
- Carry flag changed from 0 to 1

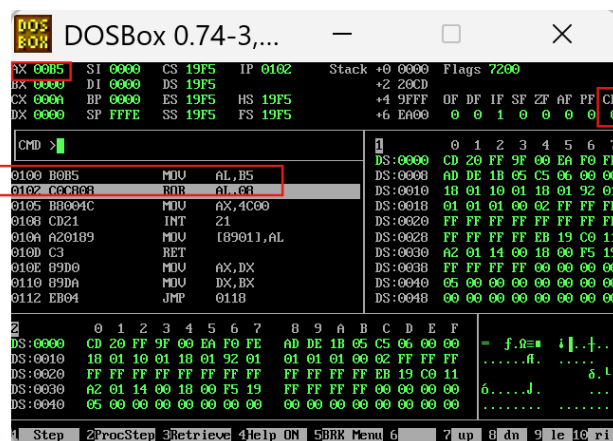
This is because:

1. LSB has been sent to MSB and all other bits are shifted right 1 time
2. LSB has been sent to carry flag

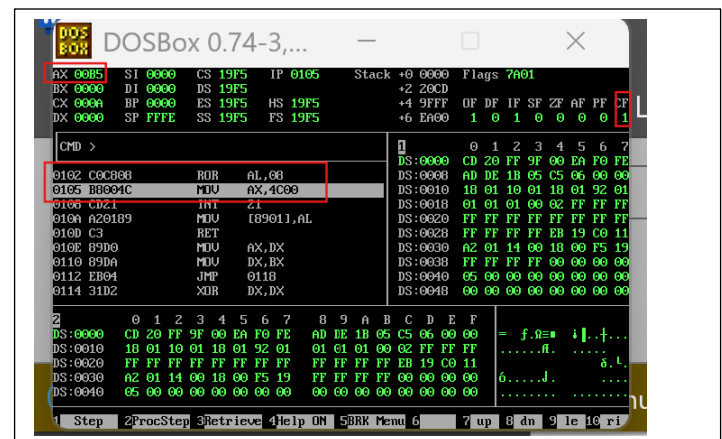
Rotate right (ROR): ROTATING 8 BIT DATA:

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | ROR al,8
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:



After shifting:



- AX has remained same B5(10110101)
- Carry flag changed from 0 to 1

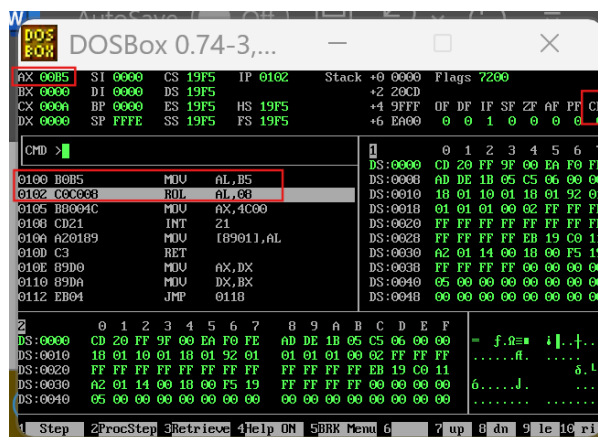
This is because:

1. With each shift LSB has been sent to MSB and as it is 8-bit data it will return to its original data
2. As the 8 bit number from left to right stores 1 carry flag is changed to 1.

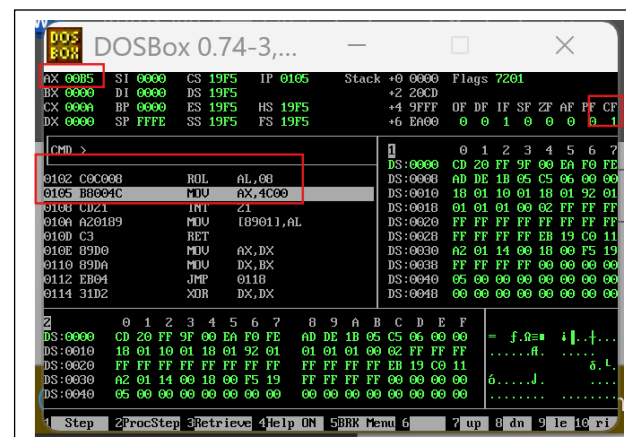
Rotate right (ROL): ROTATING 8 BIT DATA:

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | ROL al,8
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:



After shifting:



- AX has remained same B5(10110101)
- Carry flag changed from 0 to 1

This is because:

1. With each shift LSB has been sent to MSB and as it is 8-bit data it will return to its original data
2. MSB sent to carry flag

Rotate right (ROL): ROTATING 1 BIT DATA:

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | ROL al,1
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:

Register	Value
AX	00B5
SI	0000
CS	19F5
IP	0102
Stack	+0 0000
Flags	7200

Address	Instruction	Comment
0100	B0B5	MOV AL,B5
0102	D0C0	ROL AL,1
0104	B804C	MOV AX,4C00
0107	CD21	INT 21
0109	Z7	DAA
010A	A20189	MOV [8901],AL
010D	C3	RET
010E	89D0	MOV AX,DX
0110	89DA	MOV DX,BX

After shifting:

Register	Value
AX	005A
SI	0000
CS	19F5
IP	0104
Stack	+0 0000
Flags	7A01

Address	Instruction	Comment
0102	D0C0	ROL AL,1
0104	B804C	MOV AX,4C00
0107	CD21	INT 21
0109	Z7	DAA
010A	A20189	MOV [8901],AL
010D	C3	RET
010E	89D0	MOV AX,DX
0110	89DA	MOV DX,BX
0112	EB04	JMP 0110

- AX has changed from B5(10110101) to 5A(01011010)
- Carry flag changed from 0 to 1

This is because:

1. LSB has been sent to MSB and all other bits are shifted right 8 time. As it is a 8-bit number its back to its original form
2. 8th bit from the LSB has been sent to carry flag (in this case msb).

Rotate Through Carry Right (RCR): 1-bit rotation

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | RCR al,1
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:

Register	Value
AX	0005
SI	0000
DI	0000
BP	0000
SP	FFFE
CS	19F5
DS	19F5
ES	19F5
SS	19F5
FS	19F5
IP	0102
Stack	+0 0000
Flags	7200

Address	Instruction	Comment
0100	B0B5	MOV AL,B5
0102	D0B8	RCR AL,1
0104	B8004C	MOV AX,4C00
0107	CD21	INT 21
0109	Z7	DAA
010A	A201B9	MOV [01B9],AL
010D	C3	RET
010E	B9B0	MOV AX,BX
0110	B9BA	MOV DX,BX

After shifting:

Register	Value
AX	0006
SI	0000
DI	0000
BP	0000
SP	FFFE
CS	19F5
DS	19F5
ES	19F5
SS	19F5
FS	19F5
IP	0104
Stack	+0 0000
Flags	7A01

Address	Instruction	Comment
0100	B0B5	MOV AL,B5
0102	D0B8	RCR AL,1
0104	B8004C	MOV AX,4C00
0107	CD21	INT 21
0109	Z7	DAA
010A	A201B9	MOV [01B9],AL
010D	C3	RET
010E	B9B0	MOV AX,BX
0110	B9BA	MOV DX,BX
0112	B0B4	JMP 01B8

- AX has changed from B5(10110101) to 6A(01101010)
- Carry flag changed from 0 to 1

This is because:

1. Carry register initially stores 0 so it is sent to MSB
2. LSB is sent to carry register

Rotate Through Carry Right (RCR):rotating 8 times

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | RCR al,8
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:

Register	Value
AX	00B5
SI	0000
CS	19F5
IP	0102
Stack	+0 0000
Flags	7200

Address	Instruction	Comment
0100	B0B5	MOV AL,B5
0102	C0B83A	RCR AL,08
0105	B8094C	MOV AX,4C09
0108	CD21	INT 21
010A	A20189	MOV [8901],AL
010D	C3	RET
010E	89D0	MOV AX,DX
0110	89D0	MOV DX,BX
0112	EB04	JMP 0118

After shifting:

Register	Value
AX	006A
SI	0000
CS	19F5
IP	0105
Stack	+0 0000
Flags	7401

Address	Instruction	Comment
0102	C0B83A	RCR AL,08
0105	B85C4C	MOV AX,4C5C
0108	CD21	INT 21
010A	A20189	MOV [8901],AL
010D	C3	RET
010E	89D0	MOV AX,DX
0110	89D0	MOV DX,BX
0112	EB04	JMP 0118
0114	31D2	XOR DX,DX

- AX has changed from B5(10110101) to 6A(01101010)
- Carry flag changed from 0 to 1

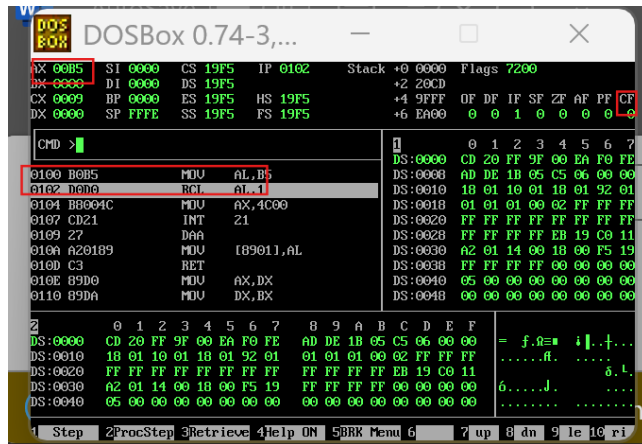
This is because:

1. Carry register initially stores 0 which is sent to MSB the first time then the LSB is sent to carry which is sent to MSB.
2. 8th bit from the LSB has been sent to carry flag (in this case msb).

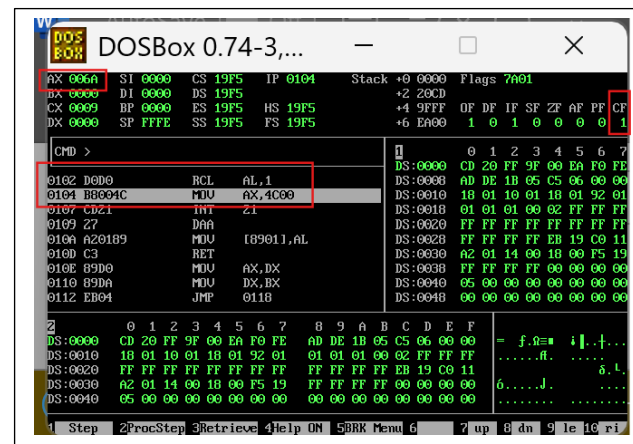
Rotate Through Carry Left (RCL): 1 bit rotation

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | rcl al,1
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:



After shifting:



- AX has changed from B5(10110101) to 6A(01101010)
- Carry flag changed from 0 to 1

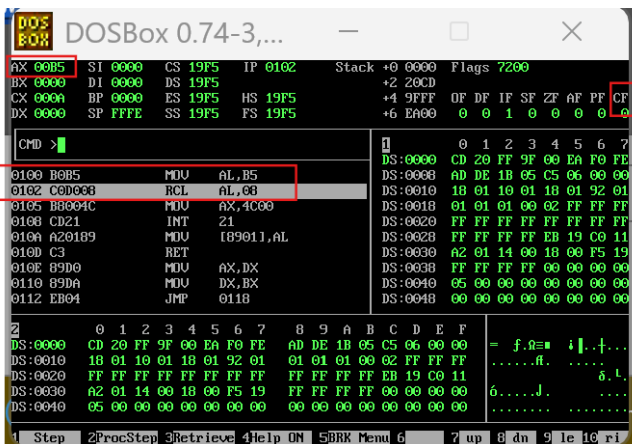
This is because:

1. Carry bit initially stores 0 which is sent to LSB .
2. Msb is sent to carry register

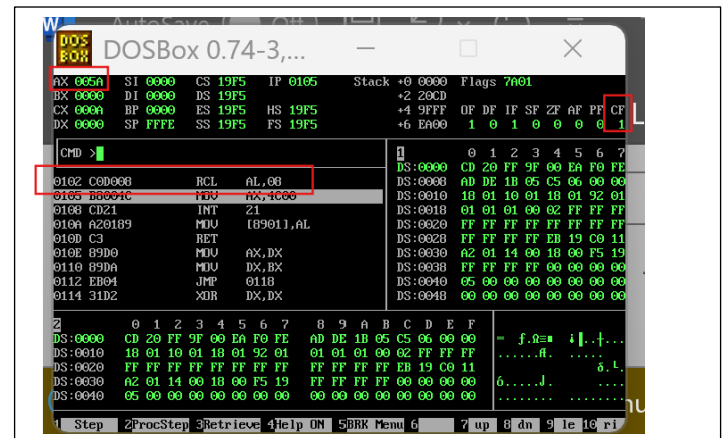
Rotate Through Carry Left (RCL): 8 bit rotation

```
1 | org 0x0100
2 |
3 | mov al,10110101b
4 |
5 | rcl al,8
6 |
7 | mov ax,0x4c00;
8 | int 0x21;
9 |
```

Before shifting:



After shifting:



- AX has changed from B5(10110101) to 5A(01011010)
- Carry flag changed from 0 to 1

This is because:

1. Carry flag initially stores 0 which is sent to LSB after which carry flag keep storing MSB which is kept sent to MSB
2. 8th bit from right side is sent to carry register